

Open Refine

for efficient data cleaning

The Goal: Get More Reliable Answers

Standardization is how we do that. Let's take a look at some FEC data: [committee disbursements to DC-area restaurants](#) (make your own copy) and find some problems of consistency. Use what you've learned from the Evaluating Data presentation to examine the recipient_name column.

- Where/how do similar values look different?
- Are there abbreviations or truncated data?
- Weird/missing punctuation or characters?

Enter some examples (and describe them) in [this spreadsheet](#).

Get Started with OpenRefine

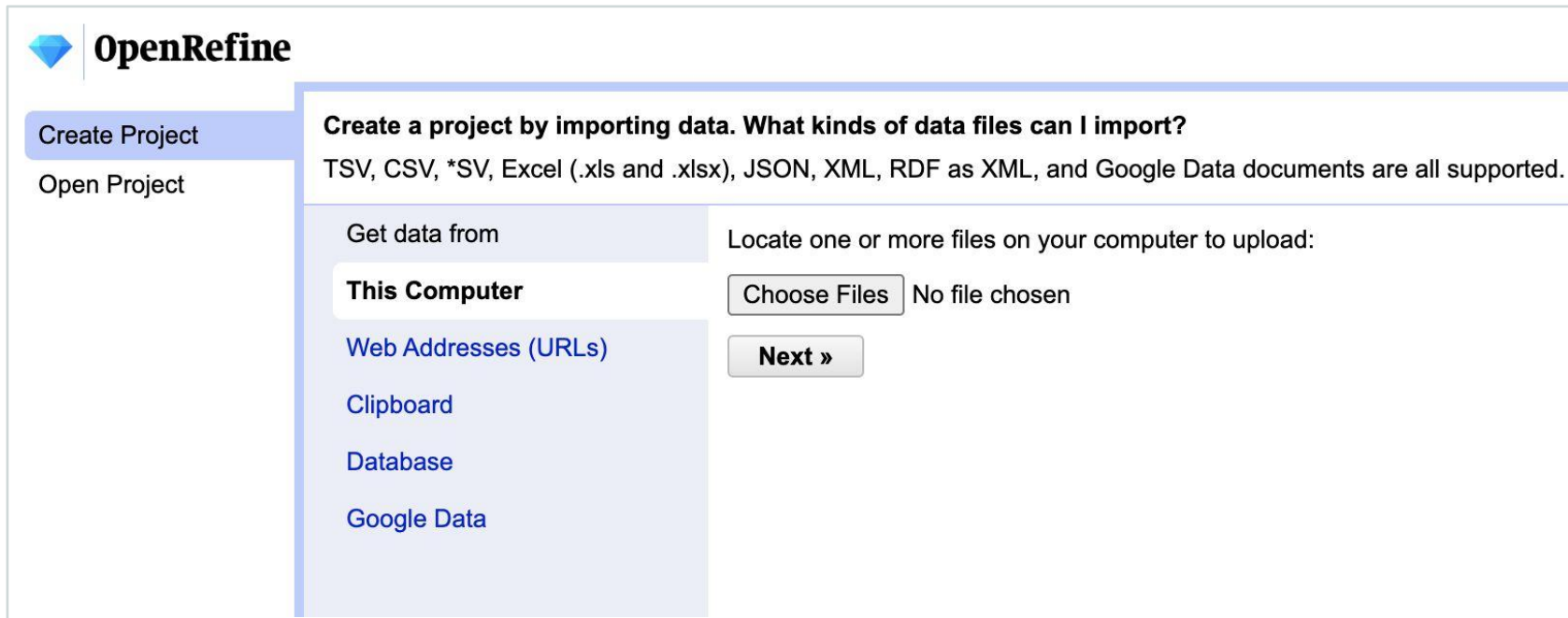
Go to the applications folder and click Open Refine to open. This should automatically launch a browser window. If it doesn't you can go here:

<http://127.0.0.1:3333/>

Then download a CSV version of the FEC restaurant-spending data from your sheet and put it in your Downloads folder.

Import the data

From the **Create Project** screen, choose **This Computer**, find your CSV, and click **Next »**.



The screenshot shows the OpenRefine web interface. On the left is a sidebar with a blue header containing the OpenRefine logo and name. Below the header are two menu items: 'Create Project' (highlighted with a blue background) and 'Open Project'. The main content area has a light blue header with the text 'Create a project by importing data. What kinds of data files can I import?' followed by a list of supported formats: TSV, CSV, *SV, Excel (.xls and .xlsx), JSON, XML, RDF as XML, and Google Data documents. Below this is a section titled 'Get data from' with a list of options: 'This Computer' (selected and highlighted in light blue), 'Web Addresses (URLs)', 'Clipboard', 'Database', and 'Google Data'. To the right of this list, the text 'Locate one or more files on your computer to upload:' is displayed above a 'Choose Files' button and the text 'No file chosen'. At the bottom right of the main content area is a 'Next »' button.

OpenRefine

Create Project
Open Project

Create a project by importing data. What kinds of data files can I import?
TSV, CSV, *SV, Excel (.xls and .xlsx), JSON, XML, RDF as XML, and Google Data documents are all supported.

Get data from

This Computer

Web Addresses (URLs)

Clipboard

Database

Google Data

Locate one or more files on your computer to upload:

Choose Files No file chosen

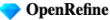
Next »

Check the Results

Once uploaded, you should see a preview. Check it out: does it look right?

Note the options in the lower-right corner of the screen. Has OpenRefine selected the right ones?

How It Should Look

 **OpenRefine** A power tool for working with messy data.

Create project

« start over

Configure parsing options

Project name Tags

Create project »

Open project

Import project

Language settings

	committee_id	committee_name	report_year	report_type	image_number	filing_form	link_id	line_number	transaction_id	file_number	entity_type	entity_type_desc	unused_recipient_committee_id	recipient_committee_id	recipient_name	recipient_ci
1.	C00558213	DEBBIE DINGELL FOR CONGRESS	2025	Q1	202504159755650501	F3	4041520251184156780	17	501227430	1887514	ORG	ORGANIZATION			A MANO	WASHINGTON
2.	C00489112	JUSTICE EQUALITY FREEDOM FOREVER PAC	2025	M2	202502109753374001	F3X	4021120251166972002	21B	XB713EB1375	1875519	ORG	ORGANIZATION			CAPITOL HILL CLUB	WASHINGTON
3.	C00489112	JUSTICE EQUALITY FREEDOM FOREVER PAC	2025	M3	202503139754145500	F3X	4031320251180009007	21B	XB721EB1380	1880230	ORG	ORGANIZATION			CAPITOL HILL CLUB	WASHINGTON
4.	C00384057	STAND WITH SANCHEZ	2025	Q1	202504159755378115	F3	4041520251184156266	17	500431686	1886648	ORG	ORGANIZATION			NATIONAL DEMOCRATIC CLUB	WASHINGTON
5.	C00832576	PATRIOT PAC	2026	M4	202604209866601489	F3X	4042020261436071146	21B	500495755	1970297	ORG	ORGANIZATION			REPUBLIC CANTINA	WASHINGTON
6.	C00443580	LUMMIS FOR WYOMING INC.	2025	Q2	202510159791358631	F3	4101520251292820003	17	B1D786C546FA04474B32	1921002	ORG	ORGANIZATION			HINE RESTAURANTS	WASHINGTON
7.	C00314575	COLLINS FOR SENATOR	2025	Q1	202506109761873638	F3	4061020251204962001	17	B983CED066DF74D1CA6B	1895008	ORG	ORGANIZATION			THE MONOCLE	WASHINGTON
8.	C00796144	VAL HOYLE FOR	2025	Q3	202510149790799870	F3	4101520251292819053	17	500343302	1919571	ORG	ORGANIZATION			AMA	WASHINGTON

Parse data as

Character encoding

Update preview

☐ Disable auto preview

CSV / TSV / separator-based files

Line-based text files

Fixed-width field text files

PC-Axis text files

JSON files

MARC files

JSON-LD files

RDF/N3 files

RDF/N-Triples files

Columns are separated by

☒ commas (CSV)

☐ tabs (TSV)

☐ custom ,

☒ Use character * to enclose cells containing column separators

☐ Trim leading & trailing whitespace from strings

Escape special characters with \

☐ Ignore first 0 line(s) at beginning of file

☒ Parse next 1 line(s) as column headers

☐ Column names (comma separated)

☐ Discard initial 0 row(s) of data

☐ Load at most 0 row(s) of data

☐ Attempt to parse cell text into numbers

☒ Store blank rows

☒ Store blank cells as nulls

☐ Store file source

☐ Store archive file

 Version 3.8.1 [TRUNK]

Preferences

Help

About

If It Looks Right

Click **Create Project** and get started. It'll look like this:

OpenRefine dc meals csv Permalink

Open... Export Help

Facet / Filter Undo / Redo 0 / 0

6715 rows

Extensions Wikibase

Show as: rows records Show: 5 10 25 50 100 500 1000 rows

« first < previous 1 next > last »

Using facets and filters

Use facets and filters to select subsets of your data to act on. Choose facet and filter methods from the menus at the top of each data column.

Not sure how to get started? Watch these screencasts

All	committee_id	committee_name	report_year	report_type	image_number	filing_form	link_id	line_number	transaction_id	file_number	entity_type	entity_type_desc	unused_recipient_committee_id
☆ 🔊	1.	C00558213	DEBBIE DINGELL FOR CONGRESS	2025	Q1	202504159755650501	F3	4041520251184156780	17	501227430	1887514	ORG	ORGANIZATION
☆ 🔊	2.	C00489112	JUSTICE EQUALITY FREEDOM FOREVER PAC	2025	M2	202502109753374001	F3X	4021120251166972002	21B	XB713EB1375	1875519	ORG	ORGANIZATION
☆ 🔊	3.	C00489112	JUSTICE EQUALITY FREEDOM FOREVER PAC	2025	M3	202503139754145500	F3X	4031320251180009007	21B	XB721EB1380	1880230	ORG	ORGANIZATION
☆ 🔊	4.	C00384057	STAND WITH SANCHEZ	2025	Q1	202504159755378115	F3	4041520251184156266	17	500431686	1886648	ORG	ORGANIZATION
☆ 🔊	5.	C00832576	PATRIOT PAC	2026	M4	202604209866601489	F3X	4042020261436071146	21B	500495755	1970297	ORG	ORGANIZATION
☆ 🔊	6.	C00443580	LUMMIS FOR WYOMING INC.	2025	Q2	202510159791358631	F3	4101520251292820003	17	B1D786C546FA04474B32	1921002	ORG	ORGANIZATION
☆ 🔊	7.	C00314575	COLLINS FOR SENATOR	2025	Q1	202506109761873638	F3	4061020251204962001	17	B983CED066DF74D1CA6B	1895008	ORG	ORGANIZATION
☆ 🔊	8.	C00796144	VAL HOYLE FOR CONGRESS	2025	Q3	202510149790799870	F3	4101520251292819053	17	500343302	1919571	ORG	ORGANIZATION
☆ 🔊	9.	C00892976	VALENTINO FOR CONGRESS	2025	TER	202503109754081978	F3	4031020251179730006	17	SB17.4380.0	1879838	ORG	ORGANIZATION
☆ 🔊	10.	C00796144	VAL HOYLE FOR CONGRESS	2025	Q3	202510149790799885	F3	4101520251292819053	17	500343323	1919571	ORG	ORGANIZATION

Facets

OpenRefine relies on things called *facets* to help us clean up data.

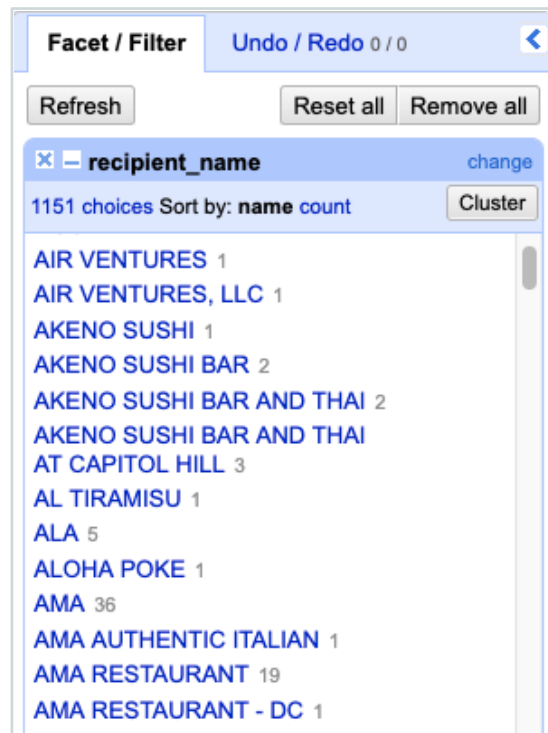
Click the triangle next to the column header `recipient_name` and select **Text facet** from the dropdown menu.

recipient_name	recipient_city	recipient_state	recipient_zip
			200072721
			20006
			20006
			200034006
REPUBLIC CANTINA	WASHINGTON	DC	200011254
HINE RESTAURANTS	WASHINGTON	DC	200032504

Facet Box

A *facet box* will appear on the left side of your workspace.

Click the **Cluster** button to use one of Refine's most powerful features.



What is Clustering?

Clustering uses different characteristics of words to group likely-identical ones together. Some techniques rely on having letters in common. Others group together words that sound alike even if they're spelled differently.

Each method has strengths and weaknesses, so it's useful to try more than one.

Putting Like Things Together

Each row here is one cluster of values
OpenRefine thinks are the same
restaurant, just typed differently.

U.S. Senate Cafe, Joe's Seafood, RPM
Italian: same payee, many spellings.
Standardizing them is what gets you
total spending per restaurant.

Cluster and edit column "recipient_name"

Find groups of different cell values that might be other representations of the same thing. For example, "New York" just differ by capitalization, and "Gödel" and "Godel" probably refer to the same person. [Find out more...](#)

Method Key collision Keying function Fingerprint

Cluster size	Row count	Values in cluster	Merge?	New cell value
3	194	- US SENATE CAFE (187 rows) - U.S. SENATE CAFE (6 rows) - U.S SENATE CAFE	☐	US SENATE CAFE
3	12	- JOE'S SEAFOOD, PRIME STEAK & STONE CRAB (9 rows) - JOE'S SEAFOOD PRIME STEAK & STONE CRAB (2 rows) - JOE'S SEAFOOD; PRIME STEAK & STONE CRAB	☐	JOE'S SEAFOOD, PRI
3	5	- FULKERSON, KENNEDY & COMPANY (3 rows) - FULKERSON KENNEDY & COMPANY - FULKERSON, KENNEDY, & COMPANY	☐	FULKERSON, KENNE
3	3	- RPM ITALIAN D.C. - RPM ITALIAN DC - RPM ITALIAN, DC	☐	RPM ITALIAN D.C.

Don't Trust, Verify

Is this a match we want? Probably! But we can check. Hover over a cluster and you get the option to **Browse this cluster**. Click it. A new window pops open showing just the rows in that potential cluster.

Here, the variants share the same city and ZIP, so they're probably the same restaurant. Close the window to return to the complete data.

Make a Copy First!

Remember how we said “make a copy of the original data”? We can do that with columns, too, so we can see the edits we make.

On the `recipient_name` column choose **Edit column**
→ **Add column based on this column...**

Name the new column something like `recipient_clean` and cluster and edit on that column.

▼ recipient_name	▼ recipient_city	▼ recipient_state	▼ recipient
Facet	ASHINGTON	DC	200072721
Text filter			
Edit cells	ASHINGTON	DC	20006
Edit column	Split into several columns...		
Transpose	Join columns...		
Sort...	Add column based on this column...		
View	Add column by fetching URLs...		
Reconcile	Add columns from reconciled values...		
REPUBLIC CANTINA	W		4
HINE RESTAURANTS	W		4
THE MONOCLE	W		7

Everyday I'm Clustering

Browse the suggested clusters.

Some look good; some won't. When one makes sense, check the **Merge?** box and confirm the **New cell value** — edit it if needed.

Then click

Merge selected & re-cluster (or **& Close**). Work through the different clustering methods.

Cluster and edit column "recipient_name"

Find groups of different cell values that might be other representations of the same thing. For example, "New York" and "new york" likely refer to the same concept and just differ by capitalization, and "Gödel" and "Godel" probably refer to the same person. [Find out more...](#)

Method: Key collision Keying function: Fingerprint ☐ Auto-update 47 clusters found

3	194	- US SENATE CAFE (187 rows) - U.S. SENATE CAFE (6 rows) - U.S SENATE CAFE	☒	U.S. SENATE CAFE
3	12	- JOE'S SEAFOOD, PRIME STEAK & STONE CRAB (9 rows) - JOE'S SEAFOOD PRIME STEAK & STONE CRAB (2 rows) - JOE'S SEAFOOD; PRIME STEAK & STONE CRAB	☒	JOE'S SEAFOOD, PRI
3	5	- FULKERSON, KENNEDY & COMPANY (3 rows) - FULKERSON KENNEDY & COMPANY - FULKERSON, KENNEDY, & COMPANY	☒	FULKERSON, KENNE
3	3	- RPM ITALIAN D.C. - RPM ITALIAN DC - RPM ITALIAN, DC	☒	RPM ITALIAN D.C.
3	17	- MEMBERS DINING ROOM (13 rows) - MEMBER'S DINING ROOM (2 rows) - MEMBERS' DINING ROOM (2 rows)	☒	MEMBER'S DINING R

Choices in cluster

Rows in cluster

Average length of choices

Length variance of choices

Select all Deselect all Export clusters Merge selected & re-cluster Merge selected & Close Close

But Wait, There's More!

OpenRefine has other useful functions worth exploring:

- Splitting cells into multiple columns
- Trimming unneeded white space
- Changing column data types
- Changing cases